

PART VI

Age and Sex Considerations in Sport and Exercise

From the previous parts of this book, we gained a solid understanding of the general principles of exercise and sport physiology. Historically, much of the basic and applied exercise physiology literature has been based on responses of young men. Now we turn our attention to how these principles are applied to children and adolescents, older individuals, and women. In [chapter 17](#), Children and Adolescents in Sport and Exercise, we examine the processes of human growth and development and how different developmental stages affect a child's physiological capacity and exercise performance. We also consider how these stages of growth and development might alter our strategies for training young athletes for competition. In [chapter 18](#), Aging in Sport and Exercise, we discuss how physiological changes affect exercise and sport performance as people age beyond the middle years, focusing on which of these changes are attributable to the aging process, per se, and what changes might be attributable to an increasingly sedentary lifestyle. We discuss the important role that continued training can play in minimizing the loss of performance capacity and physical conditioning that accompanies aging. In [chapter 19](#), Sex Differences in Sport and Exercise, we examine differences between women's and men's responses to acute exercise and exercise training and the extent to which these differences are biologically determined. We also focus on physiological and clinical issues specific to female

athletes, including menstrual function, pregnancy, osteoporosis, and the higher prevalence of eating disorders in female athletes.





17

Children and Adolescents in Sport and Exercise

In this chapter and in the web study guide

Growth, Development, and Maturation

Height and Weight

Bone

Muscle

Fat

Nervous System

ACTIVITY 17.1 *Terminology Examples* explores real-life situations that exemplify some of the key terms used in the chapter.

ACTIVITY 17.2 *Tissue Growth and Development* describes the major phases of tissue development in children and adolescents.

Physiological Responses to Acute Exercise

Strength

Cardiovascular and Respiratory Function

Metabolism

Endocrine Responses and Substrate Utilization During Exercise

AUDIO FOR FIGURE 17.9 describes the development of aerobic and anaerobic capacity in boys and girls.

ACTIVITY 17.3 *Response to Exercise and Training* considers how physiological differences in children and adolescents affect their responses to resistance, aerobic, and anaerobic training.

Physiological Adaptations to Exercise Training

Body Composition

Strength

Maximal Oxygen Consumption

Anaerobic Capacity

Physical Activity Patterns Among Youth

Children
Adolescents
Young Adulthood

Sport Performance and Specialization

Special Issues

Thermal Stress
Growth and Maturation with Training

In Closing

The fastest man and the fastest woman in the world come from the same small country of only 2.8 million people. Jamaican runners Usain Bolt and Shelly-Ann Fraser-Pryce both won track-and-field gold medals in the 100 m race in Beijing in 2008 and again in London in 2012. In 2016, Elaine Thompson won the gold in both the 100 m and 200 m races in Rio, with Bolt again winning the 100 m and 200 m races for men. How can a tiny island known for sun, beaches, and reggae music produce so many track champions? While theories are plentiful, one thing that separates Jamaican athletes is their early childhood interest in track and field, nurtured by a culture that promotes, supports, and rewards childhood exercise and sport.

Although sustaining an in-school program of regular physical activity in the United States and other countries is currently challenging, such is not the case in Jamaica. The Jamaican education system, through a rigorous physical education syllabus and dedicated physical education teachers, has paved the way for the island's Olympic track tradition. Exercise—and, in particular, running—is engrained in the culture and widely promoted among the children of the island.

Competitions encourage children to stay active, exercise regularly, and test their athletic skills against their schoolmates. As early as age 3, while children are in preschool, they begin training and preparing for one of every school's most eagerly anticipated occasions, Sports Day. Sports Days are held at virtually every school and continued throughout high school and college. Boys and girls begin to enter nationally sponsored races when they are as young as 5, and by the time they are teenagers, top sprinters are competing before huge crowds at the National Stadium at the Inter-Secondary Schools Sports Association (ISSA) Boys' and Girls' Athletic Championship, or simply "Champs," held during the first week of April every year.

Not every child will become the next Elaine Thompson or Usain Bolt, but we can learn from the Jamaican experience: Regular physical activity is good for all children, whether they grow to become world-class athletes or simply healthy and fit adults.

Throughout the previous chapters of this book, we have examined the physiological responses of adults to acute bouts of exercise and their adaptations to training and the environment. For many years, it was assumed that children and adolescents responded and adapted identically to adults, but few scientists actually had studied children and adolescents. It is especially important to understand how children and adolescents respond to exercise because physical activity is vital in battling childhood obesity and in teaching children

to develop lifelong healthy habits. We now have a better understanding and appreciation of both the differences and similarities between adults and children and adolescents, which we discuss in this chapter.

Growth, Development, and Maturation

Growth, development, and maturation are terms used to describe changes that occur in the body starting at conception and continuing through adulthood. **Growth** refers to an increase in the size of the body or any of its parts. **Development** refers to differentiation of cells along specialized lines of function (e.g., organ systems), so it reflects the functional changes that occur with growth. Finally, **maturation** refers to the process of taking on adult form and becoming fully functional, and it is defined by the system or function being considered. For example, skeletal maturity refers to having a fully developed skeletal system in which all bones have completed normal growth and ossification, whereas sexual maturity refers to having a fully functional reproductive system. The state of a child's or adolescent's maturity can be defined by

- chronological age,
- skeletal age, and
- stage of sexual maturation.

Throughout this chapter we refer to the child and the adolescent. The period of life from birth to the start of adulthood is generally divided into three phases: infancy, childhood, and adolescence. **Infancy** is defined as the first year of life. **Childhood** spans the period of time between the end of infancy (the first birthday) and the beginning of adolescence and is usually divided into early childhood (preschool) and middle childhood (elementary school). The period of **adolescence** is more difficult to define in chronological years, because it varies in both its onset and its termination. Its onset generally is defined as the onset of **puberty**, when secondary sex characteristics develop and sexual reproduction becomes possible, and its termination as the completion of growth and development processes, such as attaining adult height. For most girls,

adolescence ranges from 8 to 19 years, and for most boys from 10 to 22 years.

Given the increasing popularity of youth sport and an emphasis on increasing children's physical fitness to combat childhood obesity, we must understand the impact of exercise on the physiological aspects of growth and development. Children and adolescents must not be regarded as mere miniature versions of adults. The growth and development of their bones, muscles, nerves, and organs largely dictate their physiological and performance capacities. As a child's size increases, so do almost all of his or her functional capacities. This is true of motor ability, strength, cardiovascular and respiratory function, and aerobic and anaerobic capacity. In the following sections, we examine age-related changes in a growth and development.

Height and Weight

Specialists in the field of growth and development have spent considerable time analyzing the changes in height and weight that accompany growth and—importantly—the rates at which growth occurs over time. [Figure 17.1](#) shows that height increases rapidly during the first 2 years of life, with children reaching about 50% of adult height by age 2. After this, height increases at a progressively slower rate throughout childhood as shown by the decline in the rate of its change. Just before puberty, the rate of change in height increases markedly, followed by an exponential decrease in growth rate until full height is attained at a mean or average age of about 16 years in girls and 18 years in boys (although some boys do not reach their full height until their early 20s). The peak rate of change in height occurs at approximately 12 years in girls and 14 years in boys. The peak rate of change in body weight occurs at approximately 12.5 years in girls and 14.5 years in boys—slightly later than for height.

Bone

Bones, joints, cartilage, and ligaments form the body's structural support. Bones provide points of attachment for the muscles, protect delicate tissues, and act as reservoirs for calcium and phosphorus, and some are involved in blood cell formation. During fetal

development, as well as during the initial 14 to 22 years of life, membranes and cartilage are transformed into bone through the process of **ossification**, or bone formation. The line of cartilage in our bones is also known as the growth plate. The average age at which the growth plate closes and ossification is complete is highly variable, but bones typically begin to fuse in the preteens, and all are fused by the early 20s. On average, girls achieve full bone maturity several years before boys. This is due to the role of different hormones, including estrogen, in signaling the growth plate to close.

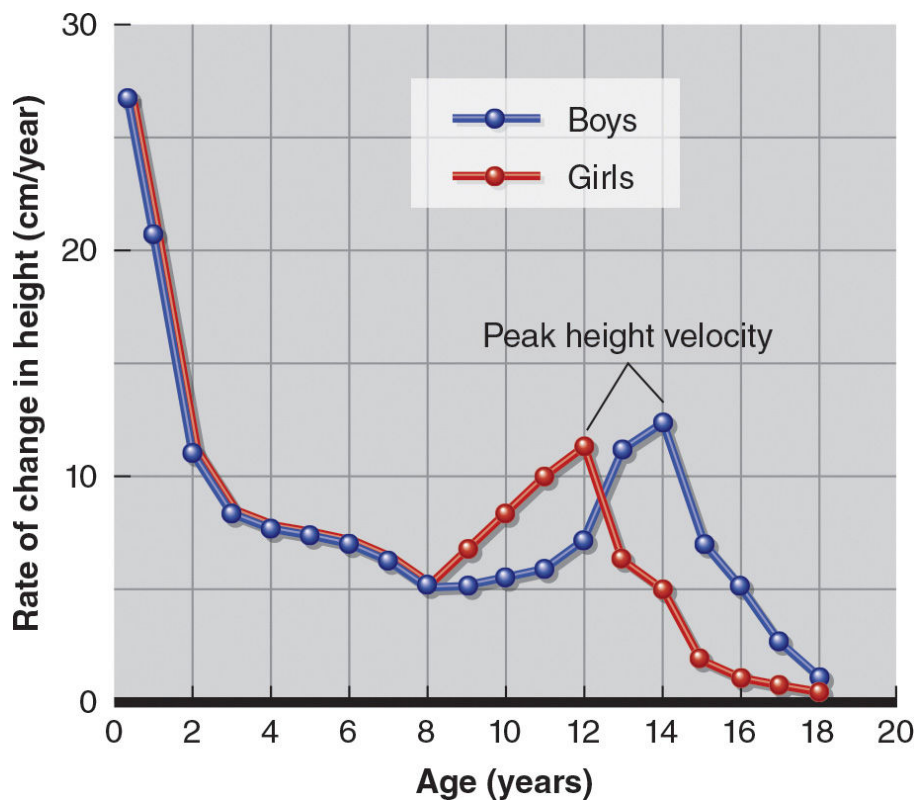


FIGURE 17.1 Changes with age in the rate of increase in height (cm/year).

In general, bone structural integrity and bone health are evaluated by examining **bone mineral density (BMD)** as well as blood markers of bone formation and resorption. During childhood and throughout adolescence, BMD increases significantly, generally peaking sometime in the second decade and falling thereafter throughout the life span. This concept is illustrated for women across the life span in [figure 17.2](#). Therefore, adolescence is a prime

window for increasing BMD with proper nutrition and physically stressing bone through weight-bearing exercise.²²

A recent longitudinal study incorporating jumping exercises (box jumping) for 8- to 9-year-old prepubescent boy and girls showed that simple short-term high-impact exercise confers long-term benefits. The boys and girls who participated in box jumping exercises during their school physical education class saw an increase in their BMD after 7 months, and this benefit was sustained for 4 years after the intervention. The increase was over and above what is observed with normal growth and development. Furthermore, if the benefits of this type of exercise are sustained until BMD plateaus in early adulthood, this type of exercise could have substantial effects on reducing fracture risks later in life when BMD decreases.¹⁰

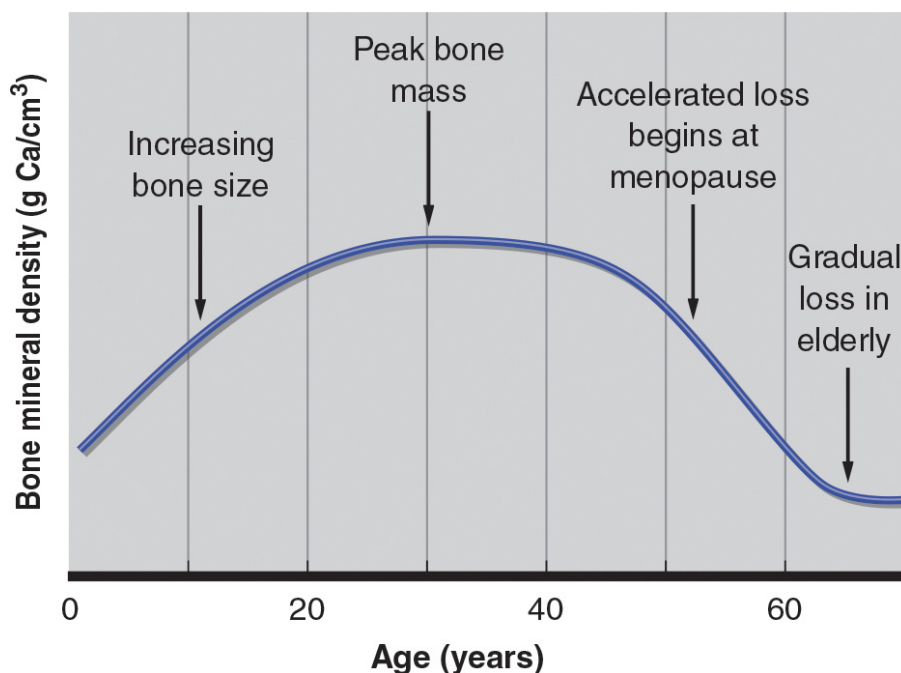


FIGURE 17.2 Changes in bone mineral density throughout the female life span. The decline after age 50 is less precipitous for men.

In Review

- Growth in height is very rapid during the first 2 years of life, with a child reaching 50% of adult stature by age 2. After that, the rate is slower throughout childhood until a marked increase occurs near puberty.

- The peak rate of height growth occurs at age 12 in girls and age 14 in boys. Full height is typically attained at age 16 in girls and age 18 in boys.
 - Growth in weight follows the same trend as height. The peak rate of weight increase occurs at age 12.5 in girls and age 14.5 in boys.
 - Bone mineral density increases significantly throughout childhood and adolescence, peaking in early adulthood. High-impact load-bearing exercise can substantially increase BMD.
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Muscle

From birth through adolescence, the body's muscle mass steadily increases as weight increases. In males, the skeletal muscle mass increases from 25% of total body weight at birth to about 40% to 45% or more in young men (20-30 years). Much of this gain occurs when the rate of muscle development peaks during puberty. This peak corresponds to a sudden, almost 10-fold increase in testosterone. Girls don't experience such rapid acceleration of muscle growth at puberty, but their muscle mass does continue to increase (although more slowly than boys') to about 30% to 35% of their total body weight as young adults. This rate difference is largely attributed to hormonal differences at puberty (see [chapter 19](#)).

Increases in muscle mass with growth appear to result primarily from hypertrophy (increase in size) of existing fibers, with little or no hyperplasia (increase in fiber number). Fiber hypertrophy results from increases in the myofilaments and myofibrils. Increases in muscle length as young bones elongate result from increases in the number of sarcomeres (which are added at the junction of the muscle and the tendon) and from increases in the length of existing sarcomeres. Muscle mass peaks in girls at age 16 to 20 years and in boys at 18 to 25 years, unless it is increased further through exercise, diet, or both.

Fat

Fat cells form and fat deposition starts in these cells early in fetal development, and this process continues indefinitely thereafter. Each fat cell can increase in size at any age from birth to death. The amount of fat that accumulates with growth and aging depends on

- diet,

- exercise habits, and
- heredity.

Heredity is unchangeable, but both diet and exercise can be altered to either increase or decrease fat stores.

At birth, 10% to 12% of total body weight is fat. At **physical maturity**, the fat content reaches approximately 15% of total body weight for men and approximately 25% for women. This sex difference, like that seen in muscle growth, is primarily attributable to hormonal differences. When girls reach puberty, their estrogen concentrations and tissue exposure increase, promoting the deposition of body fat. [Figure 17.3](#) illustrates the changes in percent body fat, fat mass, and fat-free mass for both boys and girls from ages 8 to 20 years.¹⁶ It is important to realize that both fat mass and fat-free mass increase during this time period, so an increase in absolute fat does not necessarily mean an increase in relative fat percentage.

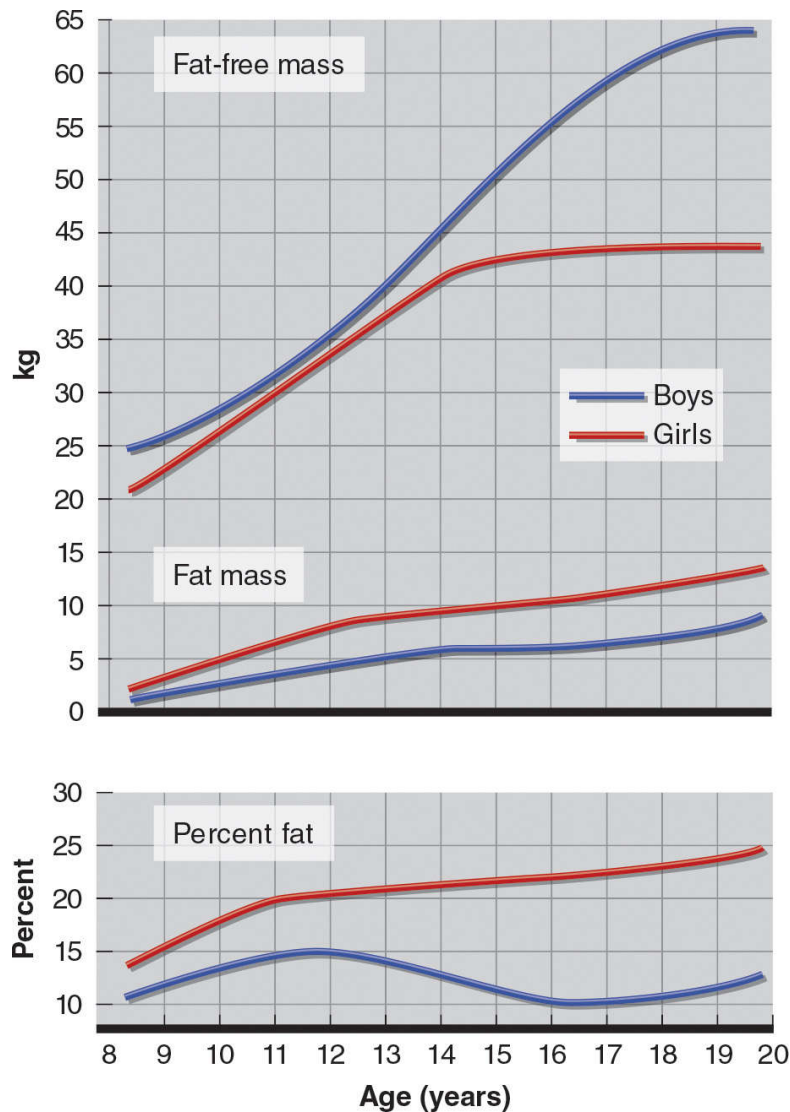


FIGURE 17.3 Changes in percent fat, fat mass, and fat-free mass for girls and boys from age 8 to 20. Reprinted by permission from R.M. Malina, C. Bouchard, and O. Bar-Or, *Growth, Maturation, and Physical Activity*, 2nd ed. (Champaign, IL: Human Kinetics, 2004), 114.

Nervous System

As children grow, they develop better balance, agility, and coordination as their nervous systems develop. Myelination of the nerve fibers must be completed before fast reactions and skilled movement can occur because conduction of an impulse along a nerve fiber is considerably slower if myelination is absent or incomplete (discussed in [chapter 3](#)). **Myelination** of the cerebral cortex occurs most rapidly during childhood but continues well beyond puberty. Although practicing an activity or skill can improve

performance to a certain extent, the full development of that activity or skill depends on full maturation (and myelination) of the nervous system. The development of strength is also likely influenced by myelination.

In Review

- Muscle mass increases steadily along with weight from birth through adolescence.
- In boys, the rate of muscle mass increase peaks at puberty, when testosterone production increases dramatically. Girls do not experience this sharp increase in muscle mass.
- Muscle mass increases in boys and girls result primarily from fiber hypertrophy with little or no hyperplasia.
- Muscle mass peaks in girls between ages 16 and 20 and in boys between ages 18 and 25, although it can be further increased through diet and exercise.
- Fat cells can increase in size and number throughout life.
- The amount of fat accumulation depends on diet, exercise habits, and heredity.
- At physical maturity, the body's fat content averages 15% in young men and 25% in young women. The differences are caused primarily by higher testosterone levels in men and higher estrogen levels in women.
- Balance, agility, and coordination improve as children's nervous systems develop.
- Myelination of nerve fibers is accompanied by faster reactions and more skilled movements because myelination speeds the transmission of electrical impulses.

Physiological Responses to Acute Exercise

The function of almost all physiological systems continues to improve until full maturity is reached, or shortly before. After that, physiological function plateaus for a period of time before starting to decline beginning in middle age. In this section, we focus on some of the changes in children and adolescents during an acute bout of exercise during growth and development.

Strength

Strength improves as muscle mass increases with growth and development. Peak strength usually is attained by age 20 in women

and between age 20 and 30 in men. The hormonal changes that accompany puberty lead to marked increases in strength in pubescent boys because of increased muscle mass. The extent of development and the performance capacity of muscle also depend on the concurrent maturation of the nervous system. High levels of strength, power, and skill are impossible if the child has not reached neural maturity. Myelination of many motor nerves is incomplete until sexual maturity, so the neural control of muscle function is limited before that time.

Figure 17.4 illustrates changes in leg strength in a group of boys from the Medford Boys' Growth Study.⁴ The boys were followed longitudinally from age 7 to 18. The rate of strength gain (slope of line) increased noticeably around age 12, the typical age for onset of puberty. Similar longitudinal data for girls over this same age span are not available. Cross-sectional data, however, indicate that girls experience a more gradual and linear increase in absolute strength and do not exhibit any significant change in strength relative to body weight after puberty,⁹ as shown in figure 17.5.

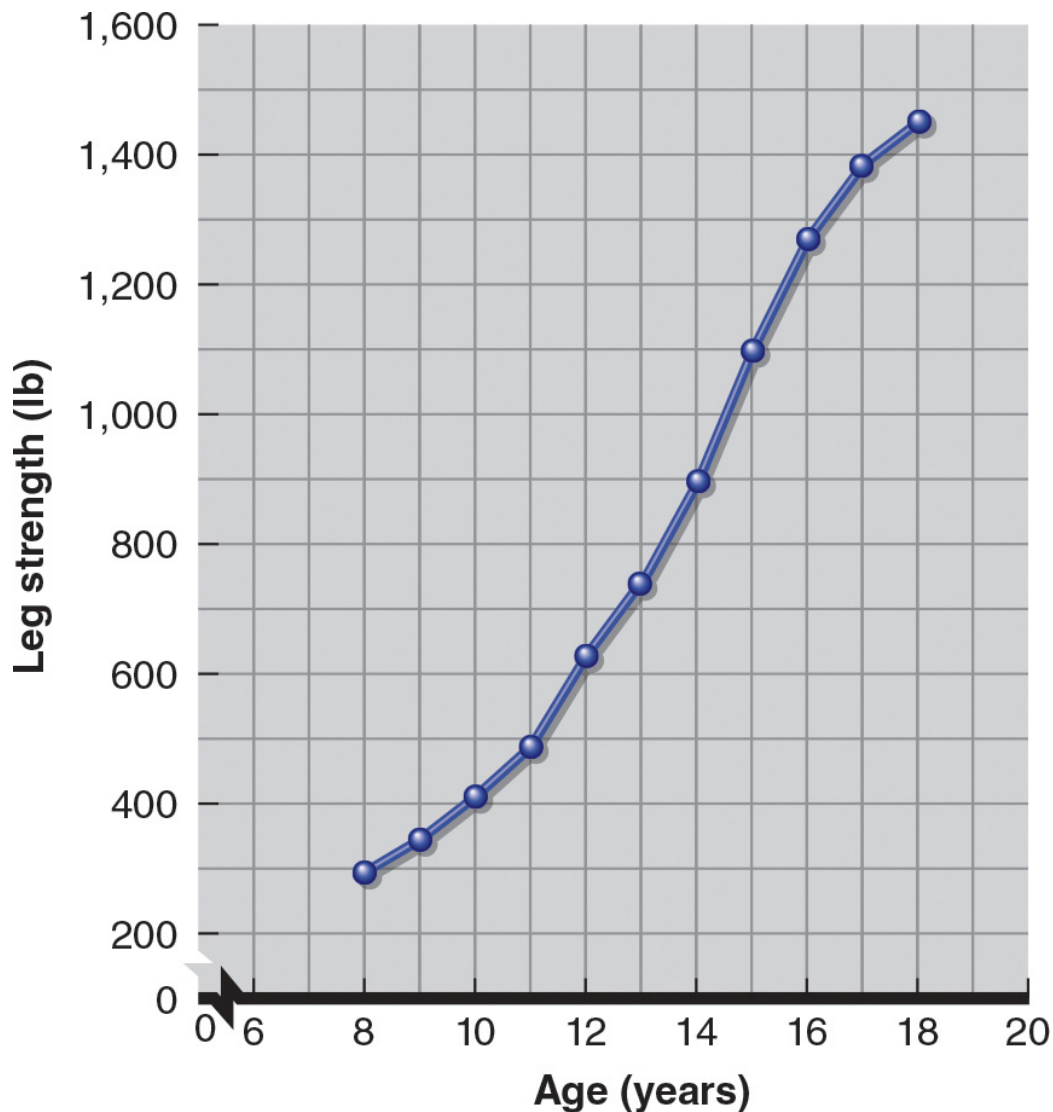


FIGURE 17.4 Gains with age in leg strength of boys followed longitudinally over 12 years. Note the increase in the slope of the curve from 12 to 16 years of age.
Data from Clarke (1971).

Cardiovascular and Respiratory Function

Cardiovascular function undergoes considerable change as children grow and age. Because of significant increases in aerobic power over the course of growth and development, we need to consider these changes at rest and during submaximal exercise, as well as during maximal exercise.

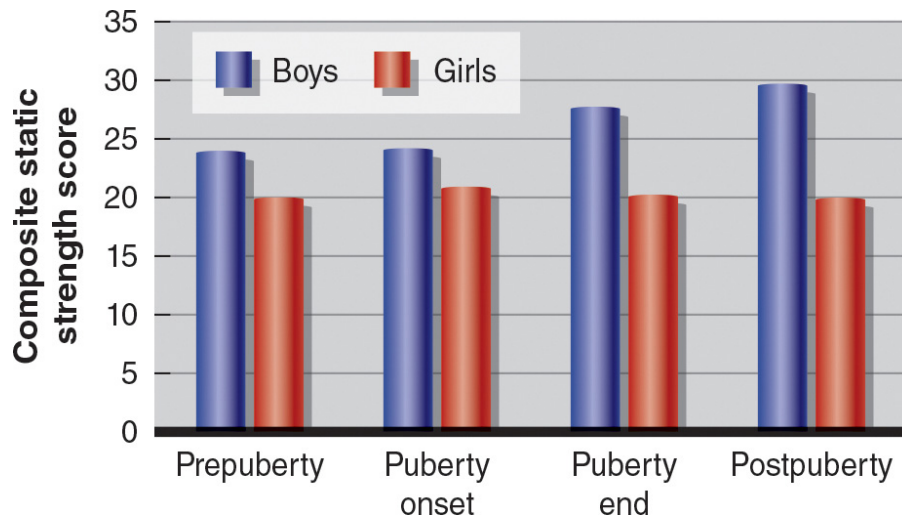


FIGURE 17.5 Changes in strength with developmental status in boys and girls. Strength is expressed as a composite static strength score from several strength testing sites, and the data are expressed per kilogram of body mass to account for differences in size between boys and girls. PHV = peak height velocity.

Rest and Submaximal Exercise

Blood pressure at rest and during submaximal exercise is lower in children than in adults but progressively increases to adult values during the late teen years. Blood pressure is directly related to body size. Larger people generally have bigger hearts and higher blood pressures, so size is at least partially responsible for children's lower blood pressure values. In addition, blood flow to active muscles during exercise in children can be greater for a given volume of muscle than in adults because children have less peripheral resistance. Thus, in children, for a given submaximal workload, blood pressure is lower and the muscle is relatively overperfused.

Recall that cardiac output is the product of heart rate and stroke volume. A child's smaller heart size and total blood volume result in a lower stroke volume, both at rest and during exercise, than in an adult. In an attempt to compensate for the lower stroke volume to maintain cardiac output, the child's heart rate at a given absolute submaximal intensity is higher than an adult's. As the child ages and heart size and blood volume increase along with body size, stroke volume increases as body size increases and heart rate decreases for the same absolute work intensity.

However, a child's higher submaximal heart rate cannot completely compensate for the lower stroke volume, so the child's

cardiac output is also somewhat lower than the adult's for a given oxygen consumption. To maintain adequate oxygen uptake during these submaximal levels of work, the child's arterial–mixed venous oxygen difference, or $(a-\bar{v})O_2$ difference, increases to further compensate for the lower cardiac output. The increase in $(a-\bar{v})O_2$ difference is most likely attributable to a greater percentage of the cardiac output going to the active muscles.²⁸ These cardiovascular responses during submaximal exercise are illustrated in [figure 17.6](#).

Maximal Exercise

Maximum heart rate ($HR_{\max}$) is higher in children than in adults but decreases linearly with age. Children under age 10 frequently have maximum heart rates exceeding 210 beats/min, whereas the average 20-year-old has a maximum heart rate of approximately 195 beats/min.

As with submaximal exercise, the child's smaller heart and blood volume limit the maximal stroke volume that he or she can achieve. Again, the high $HR_{\max}$ cannot fully compensate for this, leaving the child with a lower maximal cardiac output than the adult. This limits the child's performance at high absolute workloads during some activities (e.g., cycling) because the child's capacity for oxygen delivery is less than an adult's. However, for high relative workloads in which the child is responsible for moving only his or her body mass (e.g., running), this lower maximal cardiac output is not as serious a limitation. In running, for example, a 25 kg (55 lb) child requires (in proportion to body size) considerably less oxygen than a 90 kg (198 lb) man would require, yet the rate of oxygen consumption when scaled for body size is about the same for both.

In Review

- Strength improves in children as muscle mass increases with growth and development.
- Gains in strength with growth also depend on neural maturation because neuromuscular control is limited until myelination is complete, usually around sexual maturity.
- Blood pressure is directly related to body size: It is lower in children than in adults but increases to adult levels in the late teen years, both at rest and during

exercise.

- During both submaximal and maximal exercise, a child's smaller heart and blood volume result in a lower stroke volume than in adults. In partial compensation, a child's heart rate is higher than an adult's for the same exercise intensity. Maximal cardiac output is lower than that of an equally trained adult.
- Even with increased heart rate, a child's cardiac output remains less than an adult's. In submaximal exercise, an increase in the $(a-\bar{v})\text{O}_2$ difference ensures adequate oxygen delivery to the active muscles. But at maximal work rates, oxygen delivery limits performance in activities other than those in which the child merely needs to move body mass, such as running.

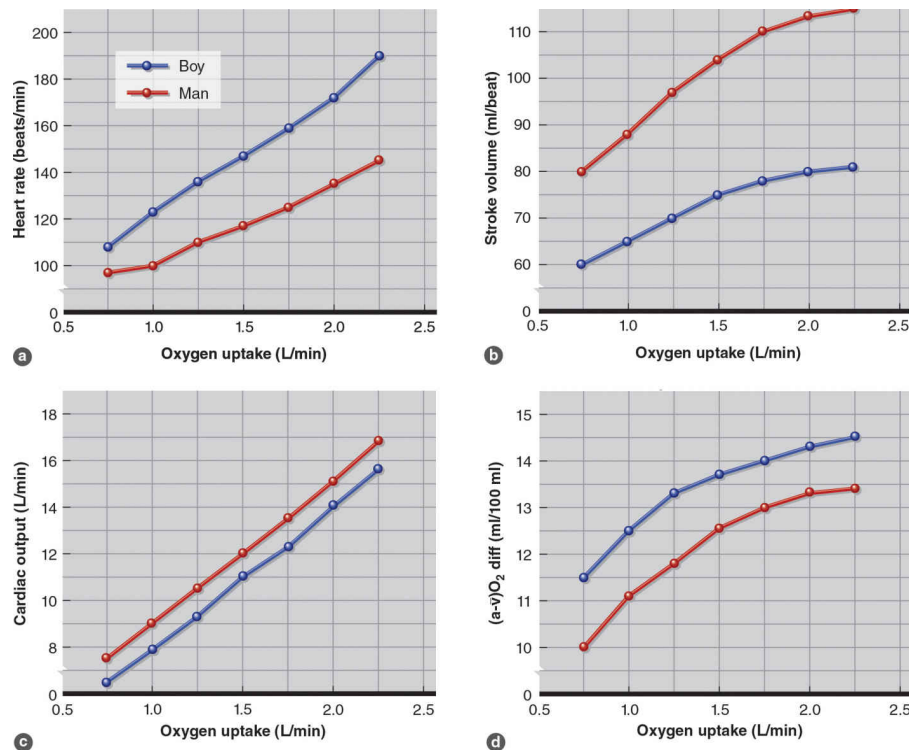


FIGURE 17.6 Submaximal (a) heart rate, (b) stroke volume, (c) cardiac output, and (d) arterial–mixed venous oxygen difference, or $(a-\bar{v})\text{O}_2$ difference, in a 12-year-old boy and a fully mature man at the same rates of oxygen uptake.

Metabolism

Metabolism and substrate utilization at rest and during exercise also change as the child and adolescent grow larger, as we would expect from the changes that we have just reviewed in muscle mass and strength and cardiorespiratory function.

Maximal Oxygen Consumption

The purpose of the cardiovascular and respiratory adjustments that occur in response to exercise is to accommodate the exercising muscles' need for oxygen. Thus, increases in cardiovascular and respiratory function that accompany growth support corresponding increases in aerobic capacity ($\dot{V}O_{2\max}$). In 1938, Robinson²⁴ demonstrated this phenomenon in a cross-sectional sample of boys and men ranging in age from 6 to 91 years. He found that $\dot{V}O_{2\max}$ peaks between ages 17 and 21 and then decreases linearly with age. Other studies have subsequently confirmed these observations. Studies of girls and women have shown essentially the same trend, although the decrease begins at a much younger age, generally age 12 to 15 years, in part attributable to an earlier assumption of a more sedentary lifestyle. The changes in $\dot{V}O_{2\max}$ with age, expressed in liters per minute, are illustrated in [figure 17.7a](#).

Scaling Physiological Data to Account for Size Differences

Expressing $\dot{V}O_{2\max}$ relative to body weight ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) provides a considerably different picture, as shown in [figure 17.7b](#). Values change little in boys from age 6 to young adulthood. For girls, little change occurs from age 6 to 13, but after age 13, $\dot{V}O_{2\max}$ shows a gradual decrease. These observations may not accurately reflect the development of the cardiorespiratory system as children grow or changes in their physical activity levels. Several concerns have been raised about the validity of using $\dot{V}O_{2\max}$ relative to body weight to account for changes in the cardiorespiratory and metabolic systems during periods of growth and development. Rather, the sex differences that begin to emerge around puberty may reflect differences in increasing body mass and changing body composition. Because girls tend to increase fat mass with estrogen exposure at puberty, decreases in $\dot{V}O_2$ per kilogram may not be significant when normalized for fat-free mass.

There are several arguments against using body weight to scale $\dot{V}O_{2\max}$ for differences in size. First, although $\dot{V}O_{2\max}$ values expressed relative to body weight remain relatively stable or decline with age, endurance performance steadily improves. The average

14-year-old boy can run the mile (1.6 km) almost twice as fast as the average 5-year-old boy, yet the two boys' $\dot{V}O_{2\max}$ values expressed relative to body weight are similar.²⁵ Second, although the increases in $\dot{V}O_{2\max}$ that accompany endurance training in children are relatively small compared with those in adults, the performance increases in these children can be larger. Therefore, body weight is likely not the most appropriate way to scale $\dot{V}O_{2\max}$ values for differences in body size in children and adolescents. The relations between $\dot{V}O_{2\max}$, body dimensions, and system functions during growth are extraordinarily complex and are discussed in greater detail later in the chapter.

Strong cases have been made for scaling $\dot{V}O_2$, cardiac output, stroke volume, and other size-related physiological variables relative to body surface area, measured in square meters, or relative to weight, expressed to the 0.67 power or the 0.75 power ($wt^{0.67}$ or $wt^{0.75}$). For years, cardiologists have expressed heart volumes relative to body surface area. Research suggests that using body surface area ($\text{ml} \cdot \text{m}^{-2} \cdot \text{min}^{-1}$) or $wt^{0.75}$ ($\text{ml} \cdot \text{kg}^{-0.75} \cdot \text{min}^{-1}$) provides the best means by which to express the data to reduce the effect of body size. One study followed young boys longitudinally from 12 to 20 years of age; one group remained untrained but active, and the other group trained.²⁷ There was little or no increase with run training in $\dot{V}O_{2\max}$ expressed in $\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$, whereas submaximal $\dot{V}O_2$ expressed in the same manner decreased with age, suggesting no change in aerobic capacity but an improvement in running economy. When these same data were expressed in $\text{ml} \cdot \text{kg}^{-0.75} \cdot \text{min}^{-1}$, the boys who were training showed increased aerobic capacity with increased training and age but no change in running economy. This latter finding intuitively makes more sense, suggesting that the use of $wt^{0.75}$ is the best way of expressing the data.

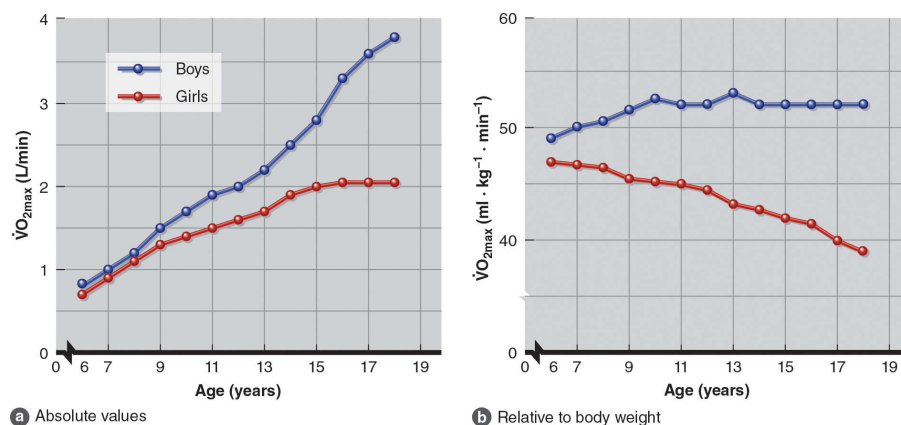


FIGURE 17.7 Changes in (a) absolute and (b) relative maximal oxygen uptake with age in children and adolescents.

Running Economy

How do growth-related changes in aerobic capacity affect a child's performance? For any activity that requires a fixed rate of work, such as cycling on an ergometer, the child's lower $\dot{V}O_{2max}$ limits endurance performance. But as noted earlier, for activities in which body weight is the major resistance to movement, such as distance running, children should not be at a disadvantage, because their $\dot{V}O_{2max}$ values expressed relative to body weight are already at or near adult values.

Yet children cannot maintain a running pace as fast as adults can because of basic differences in economy of effort. At a given speed on a treadmill, a child has a substantially higher submaximal oxygen consumption when expressed relative to body weight than an adult. As children age, their legs lengthen, their muscles become stronger, and their running skills improve. Running economy increases, and this improves their distance-running pace, even if they are not training and their $\dot{V}O_{2max}$ values don't increase.^{6,12} While an increased stride frequency as children and adolescents grow is likely the most important factor in explaining these changes in running economy, inappropriate methods of scaling oxygen consumption to body weight during growth and development, as discussed in the previous section, add to the controversy.

In Review

- As pulmonary and cardiovascular function improve with continued development, so does aerobic capacity.
 - $\dot{V}O_{2\max}$, expressed in liters per minute, peaks between ages 17 and 21 years in boys and between 12 and 15 years in girls, after which it plateaus for several years and then steadily decreases.
 - When $\dot{V}O_{2\max}$ is expressed relative to body weight, it plateaus in boys from ages 6 to 25 years before it begins to decline. In girls, the decline in $\dot{V}O_{2\max}$ is small from ages 6 to 12 years but becomes more substantial starting at about age 13. However, expressing $\dot{V}O_{2\max}$ relative to body weight might not provide an accurate estimate of aerobic capacity. Such $\dot{V}O_{2\max}$ values do not reflect the significant gains in endurance performance capacity that are noted with both maturation and training.
 - The child's lower $\dot{V}O_{2\max}$ value (L/min) limits endurance performance unless body weight is the major resistance to movement, as in distance running.
 - $\dot{V}O_{2\max}$, when expressed in liters per minute, is lower in children than in adults at similar levels of training. This is attributable primarily to the child's lower maximal cardiac output. When $\dot{V}O_{2\max}$ values are normalized for the differences in body size between children and adults, there is little or no difference in aerobic capacity, yet in activities such as distance running, a child's performance is far inferior to adult performance.
 - Running economy is lower in children compared with adults when $\dot{V}O_{2\max}$ is expressed relative to body weight. One factor has been identified to explain this difference: the difference between children and adults in stride frequency for the same fixed-pace run.
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Anaerobic Capacity

Children have a limited ability to perform anaerobic activities due to a lower glycolytic capacity. Muscle glycogen content in children is about 50% to 60% that of adults, and children do not achieve the same concentrations of lactate in either muscle or blood at maximal exercise intensities as their adult counterparts. The lower lactate concentrations might reflect a lower concentration of phosphofructokinase, the key rate-limiting enzyme of anaerobic glycolysis, and significantly lower (~3.5-fold) lactate dehydrogenase activity.¹¹ Lower blood lactate concentrations in children after exhaustive exercise may also reflect their lower relative muscle

mass, higher lactate clearance, a greater reliance on aerobic metabolism, or some combination of these. In terms of the other anaerobic metabolic pathways, children's resting stores of adenosine triphosphate (ATP) and phosphocreatine (PCr) are similar to those of adults, so activities less than 10 to 15 s in duration should not be compromised. Only activities that tax the anaerobic glycolytic system—those from 15 s up to 2 min in duration—seem to be limited.

Anaerobic mean and peak power output, as determined by the Wingate anaerobic power test (a 30 s, all-out maximal effort on a cycle ergometer), is also lower in children than in adults. [Figure 17.8](#) illustrates the results of a similar cycle ergometer anaerobic power test.²⁶ In this figure, peak power is statistically adjusted for body mass to account for differences in body size when we compare values for preteenagers, teenagers, and adults. This figure demonstrates the very low peak power outputs for preteenagers (9-10 years of age) compared with both teenagers (14-15 years of age) and adults (mean age of 21 years). Teenagers were much closer to the values for adults than the preteenagers were.

Bar-Or¹ summarized the development of both the aerobic and anaerobic characteristics of boys and girls from ages 9 through 16, using 18 years of age as the criterion for 100% of the adult value. The changes with age are shown in [figure 17.9](#). Aerobic power is represented by the child's $\dot{V}O_{2\max}$, whereas anaerobic power is represented by the child's performance on the Margaria step-running test (a field test). Maximal energy expenditure per kilogram represents the maximal energy-generating capacities of the aerobic and anaerobic systems, scaled to body weight to account for body size differences with growth. Notice that aerobic fitness remains constant for the boys but declines for the girls from 12 to 16 years. Nine- to twelve-year-old girls have a higher aerobic capacity than the eighteen-year-old reference adult value; thus, their values are 110% of the adult value. For both boys and girls, anaerobic capacity increases from 9 through 15 years of age.

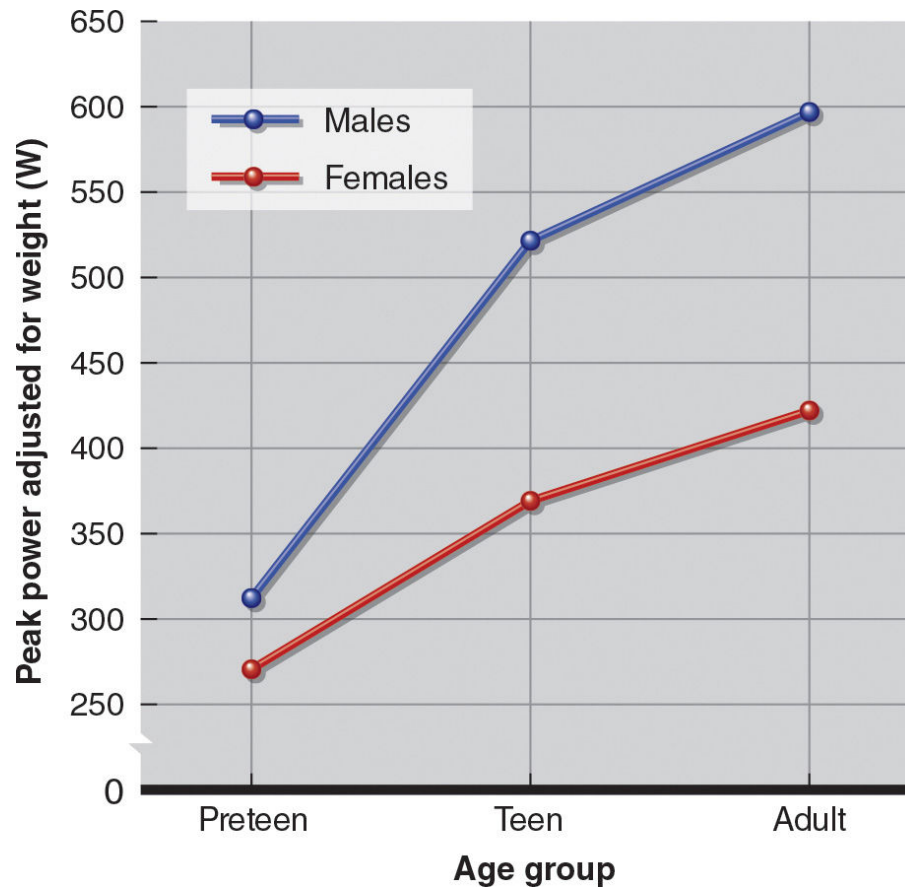


FIGURE 17.8 Optimal peak power output (anaerobic power) statistically adjusted for body mass in preteenagers (9-10 years old), teenagers (14-15 years old), and adults (mean age of 21 years). These values represent anaerobic power independent of body size.

Data from Santos et al. (2002).

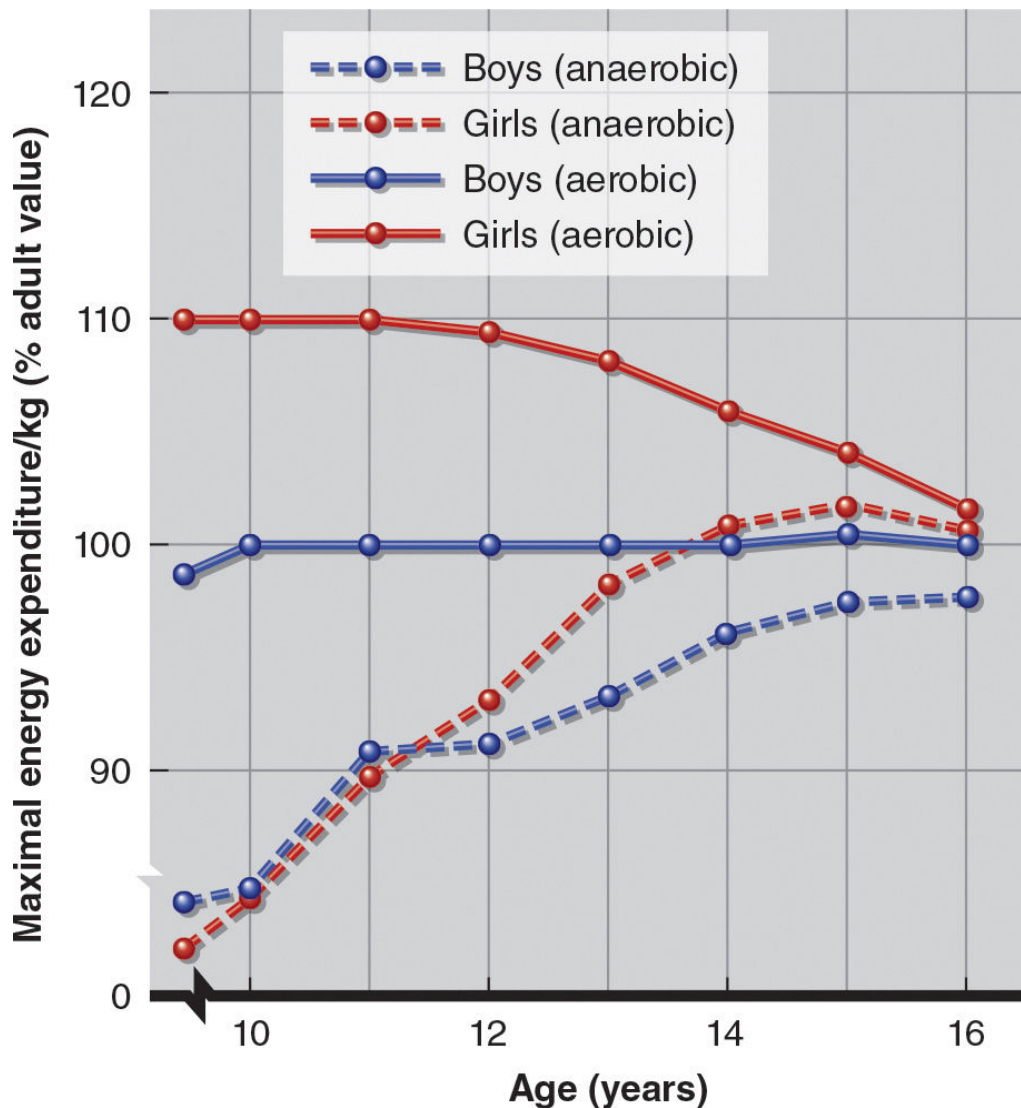


FIGURE 17.9 Development of aerobic and anaerobic characteristics in boys and girls ages 9 to 16 years. Values are expressed as a percentage of adult values (values at age 18).

Adapted by permission from O. Bar-Or, *Pediatric Sports Medicine for the Practitioner: From Physiologic Principles to Clinical Applications* (New York: Springer-Verlag, 1983). By permission of Marilyn Bar-Or.



RESEARCH PERSPECTIVE 17.1

Cognitive Benefits of Exercise for Children

Over the past decade, there has been a steady increase in classroom teaching time in an effort to improve standardized testing and academic achievement scores, involving predominately sedentary learning activities. In order to provide this increased teaching time, extensive cuts to nonacademic areas, especially those involving physical activity such as physical education and recess, have become the norm. However, a growing body of literature suggests that the exact opposite approach should likely be considered.

Physical activity/fitness and nutrition/overweight are two focus areas for *Healthy People 2020*, a comprehensive health promotion and disease prevention initiative by the Department of Health and Human Services. These are important objectives because childhood and youth physical inactivity is a growing concern and the percentage of youth considered overweight has more than tripled in the past 20 years. These statistics are alarming because of the propensity for physical inactivity and obesity to lead to chronic diseases. Schools remain an important setting in which to promote national health objectives, including lifetime physical activity. This is especially important given that physical activity habits develop very early in childhood and track over time from youth to adulthood. Thus, school programs can create expectations for regular physical activity that can persist into adulthood.

The health benefits of regular physical activity are clear. Moreover, the *Centers for Disease Control and Prevention* has published substantial evidence that physical activity can help improve academic achievement, including grades and standardized test scores. Additionally, physical activity also affects cognitive skills and attitudes as well as academic behavior, all of which improve overall academic performance, likely via improvements in concentration, attention, and classroom behavior. This scientific evidence can, and perhaps should, be used to implement policy change, gain support and funding for physical activity programs at schools and in the community, change lifestyle habits, reduce medical and financial burdens, and improve overall quality of life. Thus, physical and health education provides the foundation for healthy and active lifestyles throughout life.⁷

Endocrine Responses and Substrate Utilization During Exercise

As discussed in previous chapters, physical activity causes the release of several key metabolic regulatory hormones to mobilize carbohydrates and fats to be used as fuels. Many of the hormones that regulate metabolism during exercise can also influence growth and development. For example, exercise is a potent stimulus for the growth hormone (GH) and insulin-like growth factor axis. High-

intensity exercise in children can cause dramatic peaks in GH and influence the normal daily cycling of this hormone.

In general, pediatric studies suggest that the insulin response to exercise differs with pubertal stage and sex and that children have a higher stress response to exercise. This results in differences in blood glucose control. At the start of exercise, children have a relative hypoglycemia. The reasons for this are unclear, but in addition to lower muscle glycogen content, it is thought that children have an immature capacity for hepatic glycogenolysis. It is therefore not surprising that children rely more heavily on fat oxidation for fuel during exercise. However, exogenous glucose oxidation appears to be relatively high, potentially due to reduced endogenous glucose production. This fuel utilization profile changes throughout puberty such that adolescents have a decreased relative rate of fat oxidation, more like that in adults. The change in substrate utilization during exercise may have an impact on body composition throughout development, and from a practical point of view it may also affect nutrition needs for peak performance in children.

In Review

- Children's ability to perform anaerobic activities is limited. A child has a lower glycolytic capacity, possibly because of a limited amount of the rate-limiting enzyme phosphofructokinase or lactate dehydrogenase.
- Children have lower lactate concentrations in both blood and muscle at maximal and supramaximal rates of work.
- Anaerobic mean and peak power outputs are lower in children than in adults, even when scaled for body mass.
- Children have a different insulin and stress hormone response than adults and rely on greater fat oxidation for fuel during exercise.

Physiological Adaptations to Exercise Training

Training can improve body composition, strength, aerobic capacity, and anaerobic capacity of children. Generally, children adapt well to the same types of training regimens used by adults. But training programs for children and adolescents should be designed

specifically for each age group, keeping in mind the developmental factors associated with that age.

Body Composition

The child and adolescent respond to physical training similarly to adults with respect to changes in body weight and composition. With both resistance and aerobic training, both boys and girls decrease body weight and fat mass and increase fat-free mass, although the increase in fat-free mass is less in the child. As we have already observed, there is also evidence of significant bone growth as a result of high-impact weight-bearing exercise training,¹⁰ above that seen with normal growth.

Strength

For many years, the use of resistance training to increase muscular strength and endurance in prepubescent and adolescent boys and girls was controversial. Boys and girls were discouraged from using free weights for fear that they might injure themselves or interfere with the growth process. Furthermore, many scientists speculated that resistance training would have little or no effect on the muscles of prepubescent boys because their circulating androgen concentrations were still low. It is now widely accepted that properly designed resistance training programs are safe and effective for youth and adolescents, and the risk of injury with resistance training in youth is very low.

A number of studies conducted on both children and adolescents have clearly demonstrated that resistance training is effective in increasing strength. The increase is largely dependent on the volume and intensity of training, but in general, the percentage increases for children and adolescents are similar to those for young adults.

RESEARCH PERSPECTIVE 17.2

Physical Activity and Obesity in Children Around the World

While generic physical activity guidelines for adults were modified in 2018, global physical activity guidelines for school-aged children suggest at least 60

min/day of moderate- to vigorous-intensity activity. These guidelines are based on evidence linking physical activity to health outcomes, such as physical fitness, bone health, and markers of cardiovascular and metabolic health in children. Recently, studies have also demonstrated a negative relation between sedentary behaviors and health outcomes. Further, there is a direct association between physical activity and adiposity. However, there is a relative lack of comparative objective data on the relation between physical activity, sedentary behavior, and obesity in international samples. Thus, a group of researchers undertook a study in order to examine the association between moderate- to vigorous-intensity physical activity, vigorous physical activity, and sedentary behavior with obesity in 9- to 11-year-old children from countries that differ in stages of epidemiologic transition and represent a range of economic development (The International Study of Childhood Obesity, Lifestyle and the Environment, ISCOLE).¹²

The ISCOLE study included over 7,000 children from 12 different countries around the globe. The amount of time spent in moderate- to vigorous-intensity physical activity, vigorous physical activity, and sedentary behavior was estimated from 24 h waist-worn accelerometers used for at least 4 days. Not surprisingly, the amount of time spent in moderate- to vigorous-intensity and vigorous physical activity were significantly associated with obesity (determined from body mass index measurements). The authors report an optimal threshold of 55 min/day of moderate- to vigorous-intensity physical activity, which is consistent with the global recommendations of 60 min/day. The relation between sedentary behavior and obesity was much weaker. Taken together, these data, which were remarkably consistent between countries at different stages of economic and epidemiologic transition, strongly suggest that physical activity is directly related to obesity; this relation is evident across different cultures, races, and geographical settings. However, given the study design, it remains unclear whether low levels of physical activity are the cause or consequence of obesity in children; this remains an active area of research. Nevertheless, these data very clearly highlight a robust association between physical activity and obesity in children.

The mechanisms allowing strength changes in children are similar to those for adults, with one minor exception: Prepubescent strength gains are accomplished largely without any changes in muscle size and likely involve improvements in neural mechanisms, including improved motor skill coordination, increased motor unit activation, and other undetermined neurological adaptations.²³ Strength gains in the adolescent result primarily from both neural adaptations and increases in muscle size and specific tension.

Maximal Oxygen Consumption

Do prepubescent boys and girls benefit from aerobic training to improve their cardiorespiratory function and $\dot{V}O_{2\max}$? This also has been a controversial area because several early studies indicated that training prepubescent children did not change their $\dot{V}O_{2\max}$ values. Interestingly, even without significant increases in $\dot{V}O_{2\max}$, the running performance of the children studied did improve substantially.²⁵ Other studies have shown small increases in aerobic capacity with training in prepubescent children, but these increases are less than would be expected for adolescents or adults—about 5% to 15% in children compared with about 15% to 25% in adolescents and adults.

More substantial changes in $\dot{V}O_{2\max}$ appear to occur once children have reached puberty, although the reason for this is unknown. Because stroke volume appears to be the major limitation to aerobic performance, it is quite possible that further increases in $\dot{V}O_{2\max}$ depend on heart growth. Also, as discussed earlier in this chapter, scaling of these variables is an issue.

Anaerobic Capacity

Anaerobic training appears to improve children's anaerobic capacity. Following training, children have

- increased resting levels of PCr, ATP, and glycogen;
- increased phosphofructokinase activity;
- increased maximal blood lactate concentrations;
- and higher ventilatory thresholds.¹⁴

When designing aerobic and anaerobic training programs for children and adolescents, standard training principles for adults can be applied. Again, it is prudent to be conservative to reduce the risk of injury, overtraining, and loss of interest in sport.

In Review

- Body composition changes with training in children and adolescents are similar to those seen in adults—loss of total body weight and fat mass and increase in fat-free mass.

- The risk of injury from resistance training in young athletes is relatively low, and the programs they should follow are much like those for adults.
 - Strength gains achieved from resistance training in preadolescents result primarily from improved motor skill coordination, increased motor unit activation, and other neurological adaptations. Unlike adults, preadolescents who resistance train experience little change in muscle size. Mechanisms of strength gains in adolescents are similar to those for adults.
 - Aerobic training in preadolescents does not alter $\dot{V}O_{2\max}$ as much as would be expected for the training stimulus, possibly because $\dot{V}O_{2\max}$ depends on heart size. But endurance performance improves with aerobic training. Adolescents are similar to adults in their improvement.
 - A child's anaerobic capacity increases with anaerobic training.
-

Physical Activity Patterns Among Youth

Because physical activity patterns that are established in childhood track through adolescence and into adulthood, it is imperative to increase physical activity among our youth. However, current intervention strategies aimed at getting children more active have been mostly ineffective (discussed in the following section). [Figure 17.10](#) depicts the percentage (as of 2014) of U.S. boys and girls aged 12 to 15 who were physically active by number of days per week. Physical activity was defined as any type of moderate- to vigorous-intensity activity, both in and out of school, sufficient to increase heart rate and breathing for at least 60 min. In 2012, only about one in four U.S. children engaged in this level of activity. In the boys surveyed, as weight status increased, the percentage of boys who were active decreased significantly. A similar trend was noted among the girls, with fewer obese girls engaging in physical activity (although the difference was not statistically significant). Among active boys, basketball was the most commonly reported activity, followed by running, football, biking, and walking. Among the girls, running was the most common activity, followed by walking, basketball, dancing, and biking.⁸

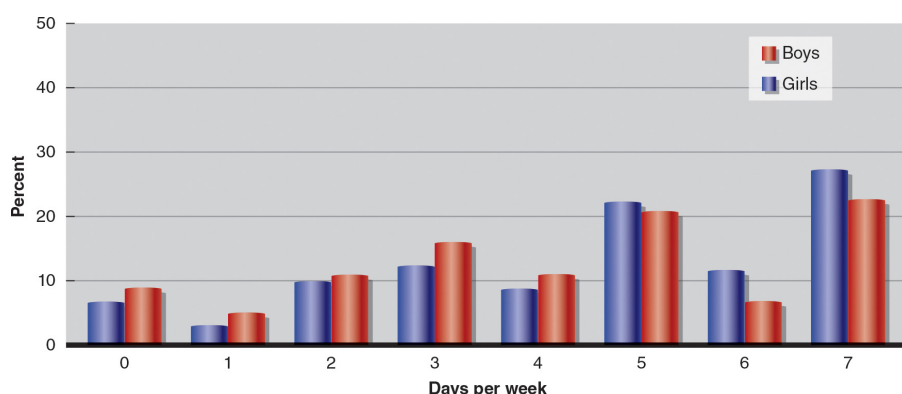


FIGURE 17.10 Percentage of U.S. boys and girls aged 12 to 15 who were physically active by number of days per week.

From U.S. Department of Health and Human Services, 2014, *HCHS Data Brief No. 141*, Available: <http://www.cdc.gov/nchs/data/databriefs/db141.htm>; Data from CDC/NCHS, National Health and Nutrition Examination Survey and National Youth Fitness Survey, 2012.

Children

Overweight children tend to be less physically active than children of normal body weight, but is physical inactivity the cause or the result of obesity? Being overweight as a child is a social and psychological impediment to physical activities that are enjoyable for normal-weight children. Therefore, for a variety of reasons, overweight children often shy away from such activities. In turn, a lack of physical activity may lead to a decrease in fitness and impaired motor skills that discourage the child even further from engaging in physical activity. It is reasonable to assume, then, that obesity in childhood may be associated with a sedentary lifestyle and poor cardiorespiratory fitness later in life.

An ongoing study in Finland²¹ is examining the association between early childhood weight status and cardiorespiratory fitness and leisure-time physical activity (LTPA) during adolescence and whether the patterns of LTPA and fitness seen during childhood persist through adolescence. This is one of the few longitudinal studies on children and adolescents in which children have been followed from birth through childhood to adolescence. A high preschool BMI was associated with low fitness in adolescence, independent of sex and regardless of adolescent LTPA. On a positive note, children who had a high preschool BMI but lost weight as they grew older had fitness levels in adolescence similar to those of the children with low BMIs. And regardless of fitness level in

childhood, if LTPA increased from age 9 to 17, a similar fitness level was seen at age 17.

The data show the importance of maintaining a healthy body weight and a physically active lifestyle from childhood through adolescence if the goal is to improve fitness during adolescence. Clearly, obesity, physical inactivity, and poor cardiorespiratory fitness in childhood and adolescence are major public health concerns, since these patterns tend to persist through adulthood.

Are Current Physical Activity Interventions for Children Working?

Children, in large numbers, fail to meet current national physical activity guidelines, despite many initiatives and interventions aimed at making our children more active. Physical activity programs have had little impact on the growing childhood obesity epidemic. According to a 2013 review and meta-analysis, the fact that current interventions to promote physical activity achieve only “a small to negligible” increase in total activity provides one explanation.¹⁷ Researchers from the United Kingdom analyzed 30 studies from the literature involving over 14,000 participants under the age of 16 years. Accelerometers, a popular method for measuring physical activity, were used to measure activity in 43% of the children studied.

The investigators looked at both total physical activity and time spent in moderate or vigorous physical activity. The disappointing results indicated that physical activity interventions have had only a small effect (about 4 min more walking or running per day) on children’s overall activity levels. Further data analysis indicated that this small intervention effect, unlikely to make any clinical difference on health variables, did not differ significantly between any of the subgroups in the study. The effect of the interventions was unaffected by age, body mass index, intervention duration, or type of program (home- or family-based intervention versus school-based intervention).



This finding may explain, in part, why physical intervention programs have had little impact on reducing the body mass index or body fat of children. The reasons that interventions have failed are less than clear. Investigators often point to flaws in the intervention designs, including poor delivery of the program, failure of the children to comply with the intervention, or lack of sufficient duration of the intervention or intensity of the exercise to elicit the designed exercise change. An alternate explanation offered by the researchers is that time spent in the intervention

may simply be replacing periods of equally intense activity. For example, after-school activity clubs may simply replace a period of time that children usually spend playing outdoors or replace a time later in the day/week when the child would usually be active. (p. 226)¹⁷

Physical Inactivity in Children and Risk of Injury

As we strive to increase physical activity in children, we must remember that unstructured free play, exercise, and organized sport all carry with them an inherent risk of injury. This risk of injury may limit some children's interest in, or enthusiasm for, healthy physical activity. It is therefore important to understand what factors increase the risk of injuries in children engaging in physical activities.

To describe the risk factors associated with injuries occurring during physical education classes, leisure-time activities, and organized sports, Bloemers and colleagues² studied a group of approximately 1,000 Dutch schoolchildren aged 9 to 12 years (the

iPlay Study). The children were followed for one school year, and physical education classes were held twice per week for 45 min per class. Activity-related injuries were continuously monitored by physical education teachers. A total of 119 injuries were reported by 104 different children over the school year. Girls were at higher risk of injury as were older children. At odds with other similar studies, there was no relation between BMI and injury risk. However, remarkably, the most active children had the lowest injury risk despite spending more time engaged in physical activity, lending support to the idea that increasing activity in children need not increase the risk of injury!

Adolescents

As described previously, childhood patterns of obesity, cardiorespiratory fitness, and physical activity portend similar patterns throughout adolescence and most likely into adulthood. Inadequate physical activity and low cardiorespiratory fitness ($\dot{V}O_{2\text{peak}}$) are clearly established as important risk factors for cardiovascular morbidity and mortality, some cancers, type 2 diabetes, and longevity, yet the relation between the two is complex. Therefore, despite the importance of understanding trends in $\dot{V}O_{2\text{peak}}$ as they relate to physical activity and other determinants, few studies have addressed whether $\dot{V}O_{2\text{peak}}$ is closely associated with physical activity during the formative adolescent years. A 2013 Norwegian study²⁰ (Young-HUNT Study) examined the distribution of cardiorespiratory fitness within a large ($n = 570$) group of healthy teenagers between the ages of 13 and 18 years. The goal was to determine the associations between $\dot{V}O_{2\text{peak}}$, self-reported physical activity, and several markers for future cardiovascular health.

A total of 289 girls and 281 boys had their resting blood pressure and heart rate, height, weight, and waist circumference measured, then underwent $\dot{V}O_{2\text{peak}}$ testing (maximal treadmill graded exercise testing). The mean $\dot{V}O_{2\text{peak}}$ was $60 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ for the boys and $49 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ for the girls. Absolute $\dot{V}O_{2\text{peak}}$ (in L/min) increased with age in both sexes. Self-reported physical activity was positively associated with $\dot{V}O_{2\text{peak}}$ for the entire group of teens,

regardless of sex or age. A high $\dot{V}O_{2\text{peak}}$ was associated with a low resting heart rate in both sexes but with a low body mass index and waist circumference only in the boys.

Adolescence is an important period of life with regard to establishing and maintaining good health habits. Although $\dot{V}O_{2\text{peak}}$ was generally high in this group of adolescents, $\dot{V}O_{2\text{peak}}$ was higher in physically active adolescents of both sexes. The strong and consistent relation between self-reported physical activity and $\dot{V}O_{2\text{peak}}$ in this study suggests that teenagers who engage in physical activity at the recommended level maintain or even increase $\dot{V}O_{2\text{peak}}$ throughout adolescence.

Young Adulthood

As adolescents transition into early adulthood, physical activity declines modestly. An analysis of 49 longitudinal studies in the literature showed that self-reported daily physical activity declined by an average of about 5 min per day.⁵ When accelerometers were used to directly measure daily activity in nine other published studies, a decline of about 7.5 min per day was seen. Both men and women had lower physical activity in adulthood compared to their late teens, but the decline was slightly larger in the men. This may be related to the fact that the men were more physically active as adolescents.

Sport Performance and Specialization

Sport performance in children and adolescents improves with growth and maturation, as can be seen for age-group records in sports such as swimming and track and field. [Figure 17.11](#) illustrates the improvement in American records for various age groups.

The figure gives values for the 100 and 400 m swim and the 100 and 1,500 m run. These events were selected because they represent a predominantly anaerobic event in swimming and running (100 m swim and run) and a predominantly aerobic activity (400 m swim and 1,500 m run). Both anaerobic and aerobic performances improve progressively with increasing age groups, with the exception of the 1,500 m run for 17- and 18-year-old girls. Similar age-group records for weightlifting do not appear to be available, because

weightlifting competition is organized by weight in broad classifications such as 16 and under, 17 to 20 years of age, and then adult classifications. On the basis of normal strength gains with growth and development, it is assumed that weightlifting records would increase markedly from late childhood through adolescence, particularly in boys.

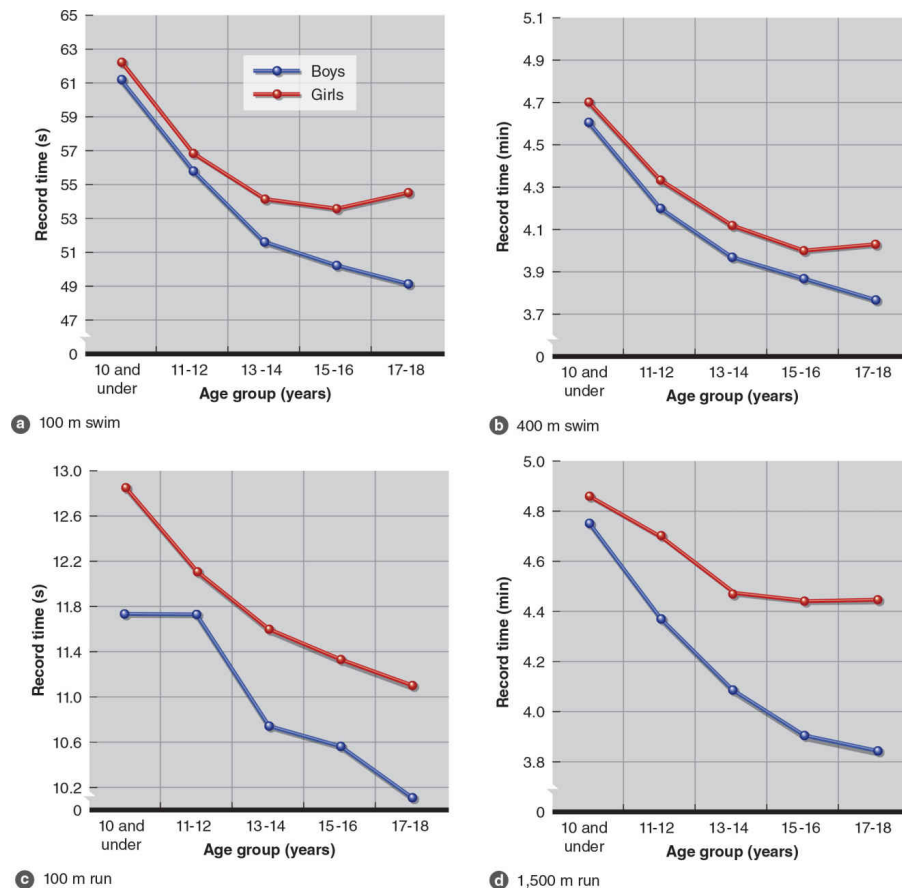


FIGURE 17.11 United States national record performances for boys and girls from 10 years of age and under through 17 to 18 years of age in (a) 100 m swim, (b) 400 m swim, (c) 100 m run, and (d) 1,500 m run.

(b) Data from USA Track & Field (2011).

As participation in youth sport grows throughout the world, so does the prevalence of year-round sport-specific training, and many children and adolescents play for multiple teams in the same sport. What began in Eastern European countries in an attempt to develop Olympic athletes has evolved into single sport-specific programs worldwide. Many parents and coaches believe that the best way to

develop their children and athletes to be elite in a sport is to have them play only that sport from an early age.¹⁴

A recent editorial¹⁹ opined that sport specialization may contribute to reduced lifelong physical activity. Reduced participation in a variety of diverse sports and fun physical activities may lead to a limited development of lifetime sport and exercise skills and interests. Furthermore, restricting exercise time to a single sport may lead to diminished overall motor skill development and increased injury risk.

The National Association for Sport and Physical Education (NASPE) has also strongly encouraged delayed sport specialization for children and adolescents. In addition to the physical benefits of engaging in multiple sports, social and psychological benefits can be seen as well. Multiple-sport athletes develop more diverse relationships and experiences than their single-sport peers, skills that are helpful in adult life. It is perhaps not surprising, then, that the most elite athletes are most commonly those who waited until later in adolescence or young adulthood to specialize in their chosen sport.

Special Issues

During the period of growth and development from childhood through adolescence, potential concerns include children's susceptibility to thermal stress and the effect of exercise training on growth and maturation. These special issues are addressed in this section.

RESEARCH PERSPECTIVE 17.3

Declines in Physical Activity During Adolescence

Physical activity declines from childhood through adulthood; however, this decline is most rapid during adolescence. Interestingly, physical activity appears to decline more rapidly in girls than in boys during this period. Given that physical inactivity is one of the most pressing public health issues, it is important to understand the underlying reasons for age- and sex-related declines in physical activity. Because adolescence is a period of rapid and dramatic change, not only in physical activity but also in physiological, psychological, and social factors, a question remains as to whether declines in physical activity are driven by biological changes or by changes in behavior and the environment.

Several studies have recently examined the adolescent fall in physical activity. One such study included data derived from the Physical Health Activity Study Team project, which ran for 5 years in the Canadian public school system and included children aged ~11 to 14 years. Participation in physical activity was quantified using a standard questionnaire. Findings from this study add to the growing body of literature documenting somewhat lower levels of overall activity and a more rapid decline in physical activity during adolescence in girls compared to boys. Interestingly, biological age (based on sexual maturity), and not chronological age (based on birth date), fully accounts for sex differences in the rate of decline but not for differences in overall levels of activity. Collectively, these findings underscore the importance of considering biological age, perhaps instead of chronological age or sex, as a main contributing factor to declines in physical activity during adolescence.

A nonintervention prospective cohort study, termed EarlyBird study, collected annual measures of accelerometry-based physical activity from ~300 children aged 5 to 15 years. The goal of the EarlyBird study was to reveal the start, extent, and nature of the decline in physical activity, determine the types of activity that contributed to the decline, and ascertain whether certain biological or environmental factors influenced the decline. The authors report that the decline in physical activity begins around 9 years of age and can be largely attributed to a reduction in light-intensity activity rather than a reduction in more intense activities.^{17,18} This may have important clinical implications for the development of intervention strategies, since light activities are more likely to be habitual whereas moderate- to vigorous-intensity physical activity is more likely to be structured. Second, the age-related decline in physical activity in both sexes was unrelated to puberty, although puberty accounted for the steeper age-related decline in physical activity among girls. Finally, physical activity appears to track throughout childhood, such that those who are inactive in early childhood are more likely to remain inactive during adolescence. In broad agreement with the Physical Health Activity Study Team project, the inactivity of adolescence is driven, at least in part, by biological variables;³ thus, these variables should be considered in the development of intervention strategies targeted to increase physical activity throughout adolescence.

Thermal Stress

Laboratory experiments suggest that children are more susceptible to heat- and cold-induced illness or injury than adults. Yet there has been a minimal number of reported cases of thermal illness or injury among children. A major concern is the child's apparently lower capacity during exercise in the heat to dissipate heat through

evaporation. Compared with adults, children have a greater ratio of body surface area to mass, meaning that they have more skin surface area from which to gain or lose heat for each kilogram of body weight. Unless the environment is hot, this is an advantage, because children are better able to lose heat through radiation, convection, and conduction. However, once the environmental temperature exceeds the skin's temperature, children more readily gain heat from the environment, which is a distinct disadvantage. A child's lower capacity for evaporative heat loss is largely the result of a lower sweating rate. Individual sweat glands in children form sweat more slowly and are less sensitive to increases in the body's core temperature than those in adults. Although young boys can acclimatize to exercise in the heat, their rate of acclimatization is slower than that of adults. Acclimatization data are not available for girls.

Only a few studies have focused on children exercising in the cold. From the limited information available, children appear to have greater conductive heat loss than adults because of a larger ratio of surface area to mass. This should be expected to place them at higher risk for hypothermia and to necessitate more clothing layers during exercise in cold temperatures.

Few studies have been conducted on children in relation to either heat or cold stress, and conclusions from existing studies sometimes have been contradictory. More research is needed in this area to determine the risks faced by children who exercise in the heat and cold. In the meantime, a conservative approach is advisable.

Growth and Maturation with Training

Many people have wondered what effect physical training has on growth and maturation. Does hard physical training slow down or accelerate normal growth and development? In a comprehensive review of this area, Malina determined that regular training has no apparent effect on growth in terms of height.¹⁵ It does, however, affect weight and body composition, as discussed earlier in this chapter.

As for maturation, the age at which peak height velocity occurs generally is not affected by regular training nor is the rate of skeletal

maturation. But the data concerning the influence of regular training on indices of sexual maturation are not as clear. Although some data suggest that menarche (the initial onset of menstruation) is delayed in highly trained girls, these data are confounded by a number of factors that generally have not been properly controlled in each study's analysis. Menarche is discussed in more detail in [chapter 19](#).

In Review

- Because physical activity patterns that are established in childhood track through adolescence and into adulthood, it is imperative to increase physical activity among our youth.
 - As more children specialize early in only one sport, the reduced participation in a variety of diverse sports and fun physical activities may lead to a limited development of lifetime sport and exercise skills and interests.
 - Laboratory studies suggest that children may be more susceptible to injury or illness from thermal stress because they have a greater ratio of body surface area to mass when compared to adults. However, the number of reported cases does not support this.
 - Children are limited in their evaporative heat loss compared to adults because children sweat less (less sweat is produced by each active sweat gland).
 - Young boys acclimatize to heat more slowly than adults do. Data are not available for girls.
 - Physical training appears to have little or no negative effect on normal growth and development. Its effects on markers of sexual maturation are less clear.
-

IN CLOSING

In this chapter, we have discussed children and young athletes. We have seen how children gain more control of movements as their body systems grow and develop, how their developing systems can sometimes limit performance capacities, and how training can improve children's performances.

We have seen that, in general, the ability to perform increases as children approach physical maturity. We have also discussed the importance of being physically active during childhood and the adolescent period, since patterns developed during these stages follow through into adulthood. Having considered the developmental process, we are now ready to consider the aging process. How is performance affected as we move beyond our physiological prime? This is our focus in the next chapter as we turn our attention to aging and the older athlete.

KEY TERMS

adolescence

bone mineral density (BMD)

childhood

development

growth

infancy

maturation

myelination

ossification

physical maturity

puberty

STUDY QUESTIONS

1. Explain the concepts of growth, development, and maturation. How do they differ?
2. At what ages do height and weight reach their peak rate of growth in boys and in girls?
3. What typical changes occur in fat cells with growth and development?
4. How does pulmonary function change with growth?
5. What changes occur in heart rate and stroke volume for a fixed rate of work as a child grows? What factors explain these changes? What changes in these two variables occur with aerobic training?

6. What changes occur in cardiac output for a fixed rate of work as a child grows? What factors explain these changes? What changes occur with aerobic training?
7. What changes occur in maximal heart rate as a child grows?
8. What physiological variables account for the $\dot{V}O_{2\max}$ increase from age 6 to 20?
9. What advice would you give to children if they wanted to improve their strength? Can they improve strength? If so, how does this occur?
10. What happens to aerobic capacity ($\dot{V}O_{2\max}$) as a prepubescent child trains aerobically?
11. What is the evidence linking childhood patterns of obesity, cardiorespiratory fitness, and physical activity with similar patterns throughout adolescence and into adulthood?
12. What happens to anaerobic capacity as a prepubescent child trains anaerobically?
13. What are the major concerns associated with early youth sport specialization?
14. How do children differ from adults with respect to thermoregulation?
15. How do physical activity and regular training affect the growth and maturation processes?

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter QUIZ tests your understanding of the material covered in the chapter.

